

Library & Collaboration Module

Scope

The **library** and **collaboration** workflows manage the controlled document lifecycle from draft review to assigned learning.

It covers:

- folder and metadata organization,
- collaboration routing (author/reviewer/approver),
- draft and approval publication to the library,
- document versioning/archive behavior,
- user/group assignment to the learning shelf,
- optional quiz question linkage for approved content.

Key Models and Objects

- Library structures: **Folder**, **File**, **Category**, **Subcategory**, **Type**, **Lab**
- Collaboration workflow: **Collaboration**, **Comment**, **Document**
- Version history: **Archived**
- Downstream assignment target: **MyShelf**, **ShelfFile**, **Question**

Role-Action Matrix

Actor	Main Actions in this module
Library Contributor (library.is_add)	Create folders/files, upload, and organize document records
Library Assigner (library.is_assign)	Assign files to users/groups (drives My Shelf workload)
Collaboration Creator (collaboration.is_add)	Create a new collaboration or reopen an existing file collaboration path
Reviewer (collaboration.is_view_and_edit)	Check out the document, update, revision, and mark review status
Approver (collaboration.is_view_and_edit)	Approve collaboration and publish the approved file into the library

Actor	Main Actions in this module
Quiz Maintainer (<code>collaboration.is_quiz_question</code>)	Add/remove quiz questions linked to collaboration document
Admin/Ops	Full override and lifecycle governance

Ordered Execution Flow

Phase 1: Library Structure and Metadata Baseline

1. Create a folder tree.
2. Configure/select category, subcategory, and type metadata.
3. Upload direct library files where collaboration is not required.

Phase 2: Collaboration Initiation

1. Creator starts collaboration flow.
2. Choose path:
 - New collaboration (new source document), or
 - Existing collaboration (reopen from an existing library file).
3. Set author, first reviewer, optional second reviewer, and approver.
4. System logs initial comment and emits invitation notifications/emails.

Phase 3: Review and Controlled Edit

1. Authorized participant checks out the collaboration file (download marks `user_downloaded`).
2. Reviewer uploads updated file revision.
3. First reviewer marks review complete.
4. Second reviewer (if configured) marks review complete after the first reviewer.
5. Comments maintain a review audit trail.

Phase 4: Approval and Publication

1. Approver marks collaboration approved when the required review chain is complete.
2. Publish action creates a finalized `File` record in the library (PDF conversion/header-footer steps for DOCX paths).
3. `last_approved`, version, and document number are persisted.
4. Draft placeholder file (if any) is removed after final publication.

Phase 5: Assignment and Consumption Handoff

1. Assigner selects target users/groups.

2. Assignment creates/updates **MyShelf** records and user linkage.
3. **ShelfFile** rows are created/updated via signal-based propagation.
4. The optional quiz question linkage follows the assigned document to the shelf users.

Quiz Workflow in Collaboration

Authoring Stage (Collaboration Context)

1. Quiz questions are added from the collaboration detail flow by users with quiz-authoring access (**collaboration.is_quiz_question**).
2. Each question stores prompt + options + correct answer in **Question**.
3. Questions are initially attached to a collaboration-linked **MyShelf** reference created in **CreateQuestionView**.
4. **get_quiz_data** and **delete_quiz_data** support listing and cleanup of collaboration quiz items before finalization.

Assignment Stage (Library to My Shelf Handoff)

1. When approved files are assigned (**save_users_to_file**), the system creates or reuses the final **MyShelf** item for that file/version.
2. Related quiz questions are re-linked to the assigned **MyShelf** item so assignees can take the quiz from the shelf.
3. Assigned users then receive the content in the **dashboard/my_shelf**, where quiz execution occurs.

Governance Note

1. Collaboration review/approval is about document quality and publication control.
2. Quiz completion is an assignee execution outcome after assignment, not an approver action.
3. For collaboration records that require learning validation, the quiz question set should be finalized before broad assignment.

Collaboration Versioning and Existing Collaboration Behavior

Versioning Rules

1. When publishing over an existing file, the current file/version is copied into **the Archive**.
2. A new approved **File** row is created with the chosen or inherited version.
3. **document_number** is preserved for existing controlled documents and generated uniquely for new document lines.

4. In the manual approve path, if the same file name + same version exists, the record is updated in place; otherwise, the old version is archived, and a new record is created.

Existing Collaboration Path

1. User selects an existing library file.
2. The system finds a prior collaboration bound to that file.
3. A new collaboration instance is created using the existing document and metadata.
4. New reviewer/approver routing is applied.
5. Collaboration goes through a full review/approval cycle again before publication.

Checkout Locking Behavior

1. File download by an authorized actor sets `user_downloaded`.
2. UI prevents parallel checkout while locked.
3. Uploading a replacement file clears the lock (`user_downloaded = None`).

Module Completion Criteria

A controlled publication cycle is complete when:

- review chain and approval are finalized,
- The library record reflects the intended version/document number,
- The previous version is archived when replaced.
- Assignment (if needed) has propagated to shelf users. When a quiz is required, quiz questions are linked to the assigned shelf item for assignee attempts.

Cross-Module Handoffs

- `dashboard/my_shelf`: assigned library files become the user workload.
- `notification`: Collaboration and assignment events emit notifications.
- `tasks` scheduler: weekly collaboration reminder emails for open collaborations.
- `account/users`: assignment and routing identities.

Execution Order Dependency Notes

Do not assign documents to learners before approval/publish completion in controlled-document workflows. Also, finalize quiz setup before broad assignment when learning validation is required, assignees can receive content without a stable quiz linkage.